

Dynamic Object Detection & Capture for UAVs

Aerial Robotics Club Summer Research

Terry Yang, Tommy Yang & Felicia Zhou

University of Toronto Institute for Aerospace Studies
Aerial Robotics Club Summer Research Program

Tuesday, August 25, 2026

Outline

- 1 Introduction
- 2 Related Works
- 3 Methodology
- 4 Results
- 5 Discussion
- 6 Acknowledgements

Problem Definition

- Unmanned aerial vehicles (UAVs) are becoming increasingly important and active within multidisciplinary applications
- **Dynamic object capture** aligns with the current trend to increase levels of autonomy and productivity in today's society
- A complex multi-stage challenge including tracking, detection, trajectory planning, interception, and capture
- Existing UAV object-interaction research often focuses on stationary objects and robotic grasping mechanisms in controlled environments

Goal and Objectives

- Build the full chain—**detect, range, frame, filter, plan**—from a single camera to a reachable catch, addressing tracking latency and true ballistic (freefall) intercept
- Use Gazebo / SIL as a **staged verification** of that loop before hardware; keep filter and planner transferable to a real flight stack
- On the aircraft: **YOLO** for detection and **PnP** for ranging; simulation may stand in with simpler vision only to close the loop

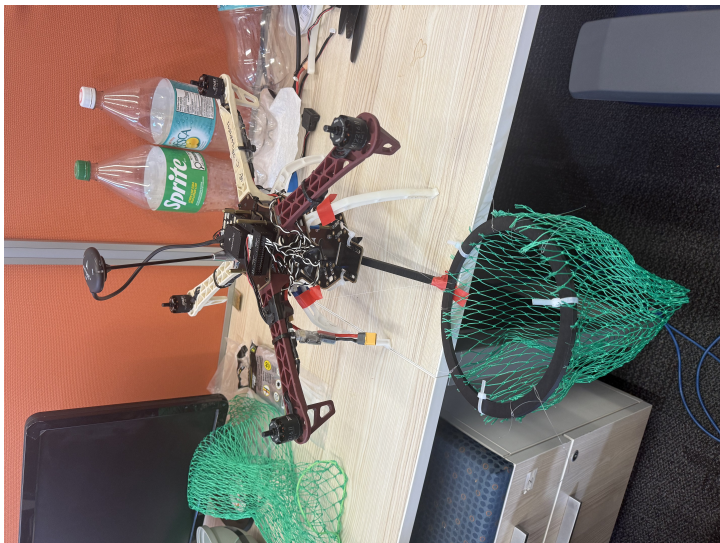
“A Real-Time Method to Detect and Track Moving Objects (DATMO) from Unmanned Aerial Vehicles (UAVs) Using a Single Camera” by Rodríguez-Canosa et al.

- Estimated camera motion is used to calculate an artificial optical flow, which is compared to the real optical flow to differentiate dynamic objects from static objects
- Complexity of recording process will contribute to detection and tracking latencies

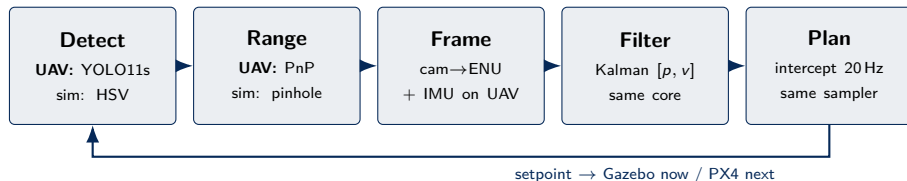
“Autonomous drone with ability to track and capture an aerial target” by García et al.

- A large net was placed on top of a frame built around the UAV to maximize the area of capture
- Ball was not in freefall which may impact orientation of the net and general structure built around the UAV

Physical Setup

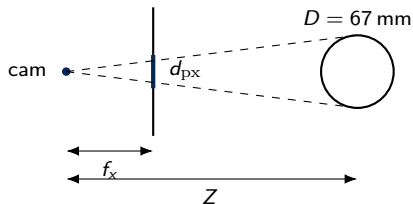


Closed-Loop Intercept Stack



Gazebo+SIL = staged close of the loop. Catch if $\|\text{net} - \text{ball}\| < 0.22 \text{ m}$,
 $z > 0.4 \text{ m}$.

Detect and Range



$$Z = 0.067 \cdot f_x / d_{px} \quad (\text{sim stand-in})$$

UAV (live path)

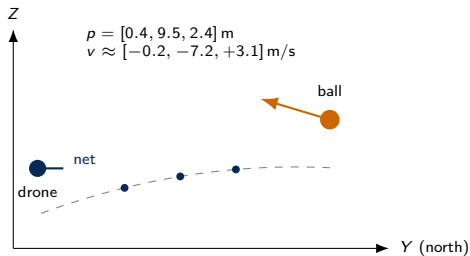
Detect: **YOLO11s** best.pt
Range: **PnP** + calibrated K

Sim (to close the loop)

Detect: yellow HSV blob
Range: pinhole diameter

YOLO/PnP already in core; Gazebo launch still
HSV+pinhole.

Filter, Plan, Throw

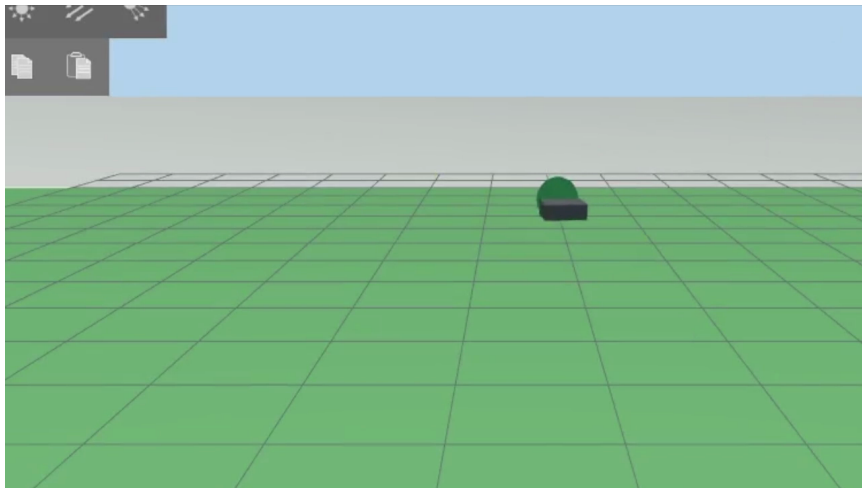


Kalman $[x, y, z, v_x, v_y, v_z]$
gravity on Z ; H reads position

Planner samples $t \in [0.18, 2.2] \text{ s}$
keep if time-to-reach $+50 \text{ ms} \leq t_{\text{go}}$

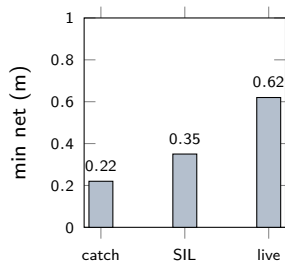
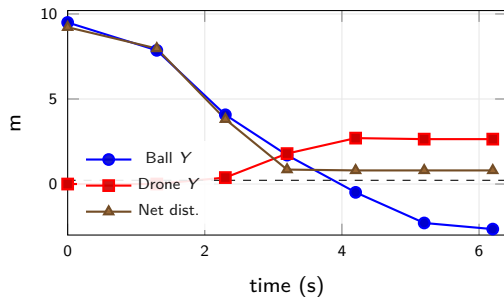
$v_{\text{max}} = 8 \text{ m/s}$, $a_{\text{max}} = 6.5 \text{ m/s}^2$

Gazebo Demo



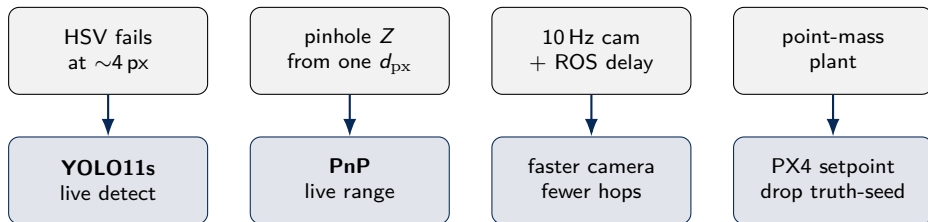
Click the frame to play gazebo_demo.mp4

Staged Sim Results



8/8 tests. SIL catches. Live
0.62–0.80 m.

From Sim Residual to Flight



Loop works at 30 Hz (SIL catch). Next: YOLO+PnP on the aircraft, same filter and planner.

- Test the system under different and more realistic environmental conditions, including wind and varying target velocities
- Develop capture mechanisms specific to certain disciplinary applications
 - Catching flying balls before they hit spectators in sports games
 - Preventing falling tools or debris on construction sites from causing head injuries
 - Mid-air retrieval of rocket when other forms of landing are infeasible

Acknowledgements

We would like to express our sincere gratitude to the ARC program at UTIAS for providing this invaluable research opportunity. We are thankful to Prof. Liu, the faculty members, and all the graduate students for their support and insightful guidance throughout the program. We are especially grateful to Kevin for his continuous mentorship and exceptional teaching; his contributions have made this experience deeply educational and truly eye-opening.

Thank You!

Questions?